EXACT COUNTEREXAMPLES AND SPECTRAL MECHANISMS FOR WOW-284

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Abstract. WOW-284 predicts that the minimum dual degree of every connected graph of order at least three and girth at least five is at most the negative of its least distance eigenvalue. The degree-seven Hoffman–Singleton graph disproves the conjecture. More generally, a degree-k Moore graph of diameter two satisfies

$$\delta^*(G) = k, \qquad \partial_{k^2+1}(G) = -\frac{3+\sqrt{4k-3}}{2}.$$

Thus WOW-284 holds for these graphs exactly when $k \le 3$, with equality at $k = 3$, and fails for every realizable $k > 3$. We give a self-contained coordinate certificate for the degree-seven example and exact induced counterexamples of orders 38, 39, 40, and 42. For every connected k-regular graph of girth at least five and diameter three we also prove

$$D = 3J + (k-3)I - 2A - A^2,$$

which converts WOW-284 into an exact shifted adjacency-spectrum condition. All finite spectral decisions use exact arithmetic. Lean 4.31 kernel-checks the complete 50-vertex proof and finite spectral certificates for the four smaller constructions.

Introduction

Let G be a finite simple connected graph on $n \ge 2$ vertices. Its distance matrix is $D(G) = (d_G(u,v))_{u,v\in V(G)}$, where d_G is graph distance. Since this matrix is real symmetric, write its eigenvalues as

$$\partial_1(G) \ge \partial_2(G) \ge \cdots \ge \partial_n(G).$$

For a vertex v, define its *dual degree* and the minimum dual degree by

$$d^*(v) = \frac{1}{d(v)} \sum_{u\in N(v)} d(u), \qquad \delta^*(G) = \min_{v\in V(G)} d^*(v).$$

Write $g(G)$ for the shortest-cycle length, with $g(G) = \infty$ for an acyclic graph; spectral superscripts indicate algebraic multiplicity.

Aouchiche and Hansen record the following Graffiti conjecture in their survey of distance spectra [1, Conjecture 7.16, p. 370]; they attribute it to Fajtlowicz's 1998 *Written on the Wall* report [2].

Conjecture (WOW-284). *If G is connected, has $n \ge 3$ vertices, and has girth $g(G) \ge 5$, then*

$$\delta^*(G) \le -\partial_n(G).$$

For such a graph, write

$$\Phi(G) := \delta^*(G) + \partial_{|V(G)|}(G).$$

Thus G is a strict counterexample precisely when $\Phi(G) > 0$.

The disproof is short. In a degree-k Moore graph of diameter two, adjacent vertices have no common neighbor and nonadjacent vertices have exactly one. If A, I, and J denote the adjacency, identity, and all-ones matrices, then

$$A^2 = (k-1)I - A + J, \qquad D = 2J - 2I - A.$$

On the orthogonal complement of the all-ones vector, the nonprincipal adjacency eigenvalues are the roots of $x^2 + x - (k-1) = 0$. It follows that

$$\partial_{k^2+1}(G) = -\frac{3+\sqrt{4k-3}}{2}.$$

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Regularity gives $\delta^*(G) = k$, so the conjectured inequality holds exactly when $k \leq 3$. The Petersen graph is the equality case $k = 3$. The degree-seven Hoffman–Singleton graph [3] has

$$\delta^* = 7, \qquad \partial_{50} = -4,$$

and therefore violates WOW-284 by the strict gap 3.

This paper develops that observation in three directions. First, a fully labelled coordinate construction verifies the Hoffman–Singleton common-neighbor identity directly, making the disproof self-contained and supplying exact graph data for independent checking. Second, natural deletions give smaller counterexamples of orders $38, 39, 40$, and 42. Third, the identity

$$D = 3J + (k-3)I - 2A - A^2$$

isolates the operator mechanism for regular diameter-three graphs and turns the conjecture into a shifted adjacency-spectrum condition. Exact characteristic polynomials, Sturm certificates, rational LDL^T decompositions, and formal-verification details are included only where they certify a stated mathematical step.

Howlader and Panigrahi provide the relevant prior spectral calculation. They determine a distance polynomial for minimal $(k, 5)$-cages and explicitly list the distance spectra of the Petersen, Hoffman–Singleton, and hypothetical degree-57 Moore graphs [4, Theorems 2.3 and 2.5(1)]; the existence of the last graph remains open [5]. Their $k = 7$ calculation, together with regularity and the girth condition, already supplies every ingredient needed to disprove WOW-284. The present note records the sharp degree criterion, makes the WOW-284 connection explicit, and supplies independent exact certificates. No claim is made that the classical distance spectra are new, that this observation has priority over every unpublished observation, that the 38-vertex example has minimum possible order among all counterexamples, or that the parameterized constructions form an unconditional infinite family.

Theorem A. *Let G be a Moore graph of diameter two and degree $k \geq 2$. Then*

$$|V(G)| = k^2 + 1, \qquad g(G) = 5, \qquad \delta^*(G) = k, \qquad \partial_{k^2+1}(G) = -\frac{3 + \sqrt{4k-3}}{2}.$$

Consequently G satisfies the inequality in WOW-284 if and only if $k \leq 3$, with equality exactly when $k = 3$.

1. The Moore-graph mechanism

A *Moore graph of diameter two and degree k* is a finite simple connected k-regular graph of diameter two attaining the Moore bound $|V(G)| \leq 1 + k + k(k-1) = k^2 + 1$. We include the short derivation needed for Theorem A; no classification of Moore graphs is used.

Fix a vertex v. There are exactly $k(k-1)$ nonbacktracking walks vuw of length two. There are also exactly $k(k-1)$ vertices outside $\{v\} \cup N(v)$, and diameter two implies that each such vertex is the endpoint of at least one of these walks. Walks with distinct endpoints are distinct, so these required walks already exhaust all $k(k-1)$ nonbacktracking walks. Consequently every such walk ends outside $\{v\} \cup N(v)$, and every outside vertex is reached exactly once. Applying this at every vertex shows that adjacent vertices have no common neighbor and nonadjacent vertices have exactly one. Thus there are no triangles or 4-cycles. To exhibit a 5-cycle, take an edge uv, choose $x \in N(u) \setminus \{v\}$ and $y \in N(v) \setminus \{u\}$, and let z be the unique common neighbor of the necessarily nonadjacent vertices x, y. The absence of triangles and 4-cycles makes these five vertices distinct, so $uxzyvu$ is a 5-cycle. Hence $g(G) = 5$.

Let A, I, and J denote the adjacency, identity, and all-ones matrices of order $n = k^2 + 1$. The common-neighbor counts give, entry by entry,

$$A^2 = (k-1)I - A + J. \tag{1}$$

Indeed, the diagonal entries are k, the entries indexed by edges are zero, and the remaining off-diagonal entries are one. Since $A\mathbf{1} = k\mathbf{1}$ and the graph is connected, the eigenvalue k is simple. For example, for $x \in \mathbb{R}^n$,

$$x^\mathsf{T}(kI - A)x = \sum_{\{u,v\} \in E(G)} (x_u - x_v)^2,$$

so equality holds only for constant x. The real symmetric matrix A preserves $\mathbf{1}^{\perp}$, and (1) restricts there to

$$A^2 + A - (k-1)I = 0.$$

Writing $q = \sqrt{4k-3}$, the two nonprincipal adjacency eigenvalues are

$$r = \frac{-1+q}{2}, \qquad s = \frac{-1-q}{2}.$$

If m_r, m_s are the respective multiplicities, then

$$m_r + m_s = k^2, \qquad k + m_r r + m_s s = 0.$$

Solving gives

$$m_r = \frac{1}{2}\left(k^2 + \frac{k(k-2)}{q}\right), \qquad m_s = \frac{1}{2}\left(k^2 - \frac{k(k-2)}{q}\right). \tag{2}$$

Every nonedge joins vertices at distance two, so

$$D = 2J - 2I - A. \tag{3}$$

It follows that the full distance spectrum is

$$\mathrm{Spec}(D) = \left\{(2k^2-k)^{(1)}, \left(\frac{q-3}{2}\right)^{(m_s)}, \left(-\frac{q+3}{2}\right)^{(m_r)}\right\}. \tag{4}$$

In particular, the last displayed eigenvalue is the least one. Regularity gives $\delta^*(G) = k$. Hence the conjectured inequality is

$$k \le \frac{3+\sqrt{4k-3}}{2}.$$

For $k \ge 2$, comparison after moving 3 to the left is legitimate, and

$$(2k-3)^2 - (4k-3) = 4(k-1)(k-3).$$

Thus the inequality is strict for $k = 2$, is an equality for $k = 3$, and fails precisely for $k > 3$. This proves Theorem A.

The smallest explicit example constructed in this note has 38 vertices. By Theorem 6.1, its minimum dual degree is $17/3$, its least distance eigenvalue is $-3-\sqrt{7}$, and its counterexample score is $8/3 - \sqrt{7} > 0$.

2. Coordinate construction

All subscripts in this section lie in the field $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. Introduce vertices

$$V(G) = \{P_{i,j} : i,j \in \mathbb{F}_5\} \mathbin{\dot\cup} \{Q_{k,\ell} : k,\ell \in \mathbb{F}_5\}.$$

Thus $|V(G)| = 50$. The edges are the following unordered pairs:

$$E_P = \big\{\{P_{i,j}, P_{i,j+1}\} : i,j \in \mathbb{F}_5\big\},$$
$$E_Q = \big\{\{Q_{k,\ell}, Q_{k,\ell+2}\} : k,\ell \in \mathbb{F}_5\big\},$$
$$E_X = \big\{\{P_{i,j}, Q_{k,ik+j}\} : i,j,k \in \mathbb{F}_5\big\}.$$

Set $E(G) = E_P \cup E_Q \cup E_X$. For a fixed numerical labeling, with representatives in $\{0,1,2,3,4\}$, use

$$P_{i,j} \longleftrightarrow 5i + j, \qquad Q_{k,\ell} \longleftrightarrow 25 + 5k + \ell.$$

After reindexing the Q-layers by $k \mapsto -k$, these are exactly Hafner's affine-coordinate form of Robertson's pentagon-and-pentagram construction [6, Section 2 and Theorem 2.1]. Thus G is the Hoffman–Singleton graph; all properties needed here are proved directly.

The complete neighborhood formulas are

$$N(P_{i,j}) = \{P_{i,j-1}, P_{i,j+1}\} \cup \{Q_{k,ik+j} : k \in \mathbb{F}_5\}, \tag{5}$$
$$N(Q_{k,\ell}) = \{Q_{k,\ell-2}, Q_{k,\ell+2}\} \cup \{P_{i,\ell-ik} : i \in \mathbb{F}_5\}. \tag{6}$$

Lemma 2.1. *The construction defines a simple 7-regular graph with 175 edges.*

Proof. Every listed edge has distinct endpoints. Within each of the five fixed P-layers, addition by 1 gives a 5-cycle; within each fixed Q-layer, addition by 2 gives a 5-cycle. The cross edges have one endpoint of each type. The set notation removes any possibility of parallel edges.

Equations (5)–(6) list seven distinct neighbors at every vertex. For a P-vertex, the two same-layer neighbors are distinct because $1 \neq -1$ in $\mathbb{F}_5$; the five cross neighbors are distinct because their first Q-coordinates are distinct; and the two types cannot overlap. The Q-case is identical with steps ± 2. There are 25 edges of type E_P, 25 of type E_Q, and 125 cross edges, hence 175 in total. Equivalently, the handshake identity gives $50 \cdot 7/2 = 175$. □

3. The common-neighbor certificate

Proposition 3.1. *For any two distinct vertices x, y,*

$$|N(x) \cap N(y)| = \begin{cases} 0, & x \sim y, \\ 1, & x \nsim y. \end{cases}$$

Proof. First consider two P-vertices $P_{i,j}$ and $P_{i',j'}$. If $i = i'$, their layer is a 5-cycle. Adjacent pairs have no common P-neighbor. If, for example, $j' = j + 2$, the unique common P-neighbor is $P_{i,j+1}$; the case $j' = j - 2$ is analogous. They cannot have a common Q-neighbor: for fixed i and k, the equation $Q_{k,\ell} \sim P_{i,j}$ determines $j = \ell - ik$ uniquely. If $i \neq i'$, there is no common P-neighbor, and a common Q-neighbor must satisfy

$$ik + j = i'k + j'.$$

This has the unique solution

$$k = \frac{j' - j}{i - i'}, \qquad \ell = ik + j,$$

because $\mathbb{F}_5$ is a field.

The argument for two Q-vertices is dual. If their first coordinates agree, they lie in the 5-cycle generated by steps ± 2. Adjacent pairs have no common Q-neighbor. For a nonadjacent pair, whose second coordinates differ by ± 1, direct intersection of the two sets of ± 2-neighbors gives exactly one common Q-neighbor; no common P-neighbor is possible. If $k \neq k'$, a common $P_{i,j}$ is determined uniquely by

$$\ell = ik + j, \qquad \ell' = ik' + j,$$

namely

$$i = \frac{\ell - \ell'}{k - k'}, \qquad j = \ell - ik.$$

It remains to compare $P_{i,j}$ with $Q_{k,\ell}$. Put

$$r = \ell - (ik + j) \in \mathbb{F}_5.$$

The pair is adjacent precisely when $r = 0$. A common P-vertex must be $P_{i,j+s}$ with $s \in \{\pm 1\}$, and it is adjacent to $Q_{k,\ell}$ precisely when $r = s$. Hence there is exactly one common P-neighbor if $r \in \{\pm 1\}$, and none otherwise. A common Q-vertex must be $Q_{k,\ell+t}$ with $t \in \{\pm 2\}$, and it is adjacent to $P_{i,j}$ precisely when $t = -r$. Hence there is exactly one common Q-neighbor if $r \in \{\pm 2\}$, and none otherwise. Finally,

$$\mathbb{F}_5 = \{0\} \,\dot\cup\, \{\pm 1\} \,\dot\cup\, \{\pm 2\},$$

which proves every cross case and completes the certificate. □

Corollary 3.2. *The graph is connected, has diameter two, and has girth five.*

Proof. Every pair of distinct vertices is adjacent or has a common neighbor, so the graph is connected and has diameter at most two. It is not complete, since it is 7-regular on 50 vertices, so its diameter is exactly two.

A triangle would give an adjacent pair a common neighbor. In a 4-cycle, two opposite vertices would have the other two cycle vertices as distinct common neighbors. Both possibilities contradict Proposition 3.1. Thus $g(G) \geq 5$. On the other hand, for every fixed i,

$$P_{i,0} P_{i,1} P_{i,2} P_{i,3} P_{i,4} P_{i,0}$$

is a 5-cycle, so $g(G) = 5$. □

4. The 50-vertex coordinate counterexample

The graph just constructed is 7-regular on $50 = 7^2 + 1$ vertices and has diameter two. It is therefore a Moore graph to which Theorem A applies. Substituting $k = 7$ into (4) gives

$$\operatorname{Spec}(D) = \{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}, \tag{7}$$

or equivalently

$$\det(tI - D) = (t-91)(t-1)^{21}(t+4)^{28}.$$

Every vertex and every one of its neighbors has degree 7, so

$$d^*(v) = \frac{1}{7}\sum_{u\in N(v)} 7 = 7.$$

Consequently

$$g(G) = 5, \qquad \delta^*(G) = 7 > 4 = -\partial_{50}(G), \qquad \delta^*(G) + \partial_{50}(G) = 3.$$

This proves that the explicitly constructed Hoffman–Singleton graph is a strict counterexample to WOW-284.

Equivalently, $D + \delta^*(G)I = D + 7I$ has eigenvalues 98, 8 with multiplicity 21, and 3 with multiplicity 28. Thus it is positive definite, which is the matrix form of the strict inequality $\delta^*(G) > -\partial_{50}(G)$.

5. A 40-vertex induced counterexample

The same coordinates contain a smaller regular counterexample. Delete

$$\mathcal{P} = \{P_{0,j}, Q_{0,j} : j \in \mathbb{F}_5\}$$

and let $R = G - \mathcal{P}$ be the induced graph on the remaining 40 vertices. The deleted graph consists of the $P_{0,\bullet}$ pentagon, the $Q_{0,\bullet}$ pentagram, and the matching $P_{0,j}Q_{0,j}$; hence it is a Petersen graph. This construction is the classical 40-vertex $(6,5)$-cage of O'Keefe and Wong [7]. Its uniqueness was proved by Wong [8], and its realization as a Petersen deletion from the Hoffman–Singleton graph is recorded explicitly by Klin, Muzychuk, and Ziv-Av [9, Section 3.6].

Theorem 5.1. *The graph R is connected and 6-regular, has girth five and diameter three, and has distance spectrum*

$$\operatorname{Spec}(D(R)) = \{75^{(1)}, 3^{(5)}, 0^{(16)}, (-5)^{(18)}\}.$$

Consequently

$$\delta^*(R) = 6 > 5 = -\partial_{40}(R), \qquad \delta^*(R) + \partial_{40}(R) = 1,$$

so R is a strict counterexample to WOW-284.

Proof. Every retained vertex has exactly one neighbor in $\mathcal{P}$: $P_{i,j}$, with $i \neq 0$, is adjacent there only to $Q_{0,j}$, while $Q_{k,\ell}$, with $k \neq 0$, is adjacent there only to $P_{0,\ell}$. Thus R is 6-regular. It inherits the absence of triangles and 4-cycles from G, and each retained P-layer is a 5-cycle, so $g(R) = 5$.

Fix $v \in V(R)$. Girth five makes the distance-zero, distance-one, and distance-two layers from v have respectively 1, 6, and $6 \cdot 5 = 30$ vertices. Three vertices remain. Each has no neighbor in $L_0(v) \cup L_1(v)$ and has at most two neighbors among the three remaining vertices; 6-regularity therefore gives it a neighbor in the distance-two layer. Hence all three lie at distance three from v. This proves connectedness and diameter three.

It remains to compute the spectrum. Order the vertices of G with those of R first, and write its adjacency matrix in blocks as

$$A_G = \begin{pmatrix} B & C \\ C^{\mathsf{T}} & P \end{pmatrix},$$

where $B = A(R)$, $P = A(G[\mathcal{P}])$, and C is 40×10. Every row of C has sum one and every column has sum four. Moreover, direct inspection of the two 5-cycles and their matching shows that every nonadjacent pair of vertices in $\mathcal{P}$ has exactly one common neighbor in $\mathcal{P}$. Consequently, distinct vertices of $\mathcal{P}$ have no common retained neighbor: adjacent pairs have no common neighbor in G, while the unique common neighbor of a nonadjacent pair already lies in the induced Petersen graph. Therefore

$$C\mathbf{1}_{10} = \mathbf{1}_{40}, \qquad C^{\mathsf{T}}C = 4I_{10}. \tag{8}$$

In particular, C has rank ten.

Equation (1) with $k = 7$ gives $A_G^2 = 6I - A_G + J$. Its top-right block is

$$BC + CP = -C + J_{40,10}.$$

Here $J_{a,b}$ denotes the $a \times b$ all-ones matrix. Since $J_{40,10} = CJ_{10}$ by (8),

$$BC = C(J_{10} - I_{10} - P). \tag{9}$$

The Petersen adjacency spectrum, obtained from the $k = 3$ case of Section 1, is $\{3^{(1)}, 1^{(5)}, (-2)^{(4)}\}$. Because C is injective, equation (9) gives

$$\operatorname{Spec}(B|_{\operatorname{col} C}) = \{6^{(1)}, (-2)^{(5)}, 1^{(4)}\}.$$

The top-left block of the same identity is

$$B^2 + CC^{\mathsf{T}} = 6I_{40} - B + J_{40}.$$

Since B is symmetric, $\ker C^{\mathsf{T}} = (\operatorname{col} C)^{\perp}$ is B-invariant. If $x \in \ker C^{\mathsf{T}}$, then $\mathbf{1}_{40}^{\mathsf{T}} x = (C\mathbf{1}_{10})^{\mathsf{T}} x = 0$, so $(B^2 + B - 6I)x = 0$. Thus the remaining 30 adjacency eigenvalues are 2 or -3. If their multiplicities are a, b, then $a + b = 30$; the displayed ten-dimensional spectrum has trace zero, and $\operatorname{tr} B = 0$, so $2a - 3b = 0$. Hence $a = 18$, $b = 12$, and

$$\operatorname{Spec}(B) = \{6^{(1)}, 2^{(18)}, 1^{(4)}, (-2)^{(5)}, (-3)^{(12)}\}. \tag{10}$$

Let A_i be the distance-i matrix of R. Girth five and 6-regularity give $A_2 = B^2 - 6I$, while diameter three gives $J = I + B + A_2 + A_3$. Consequently

$$D(R) = B + 2A_2 + 3A_3 = 3J + 3I - 2B - B^2. \tag{11}$$

On $\mathbf{1}_{40}$, the eigenvalue is $3 \cdot 40 + 3 - 2 \cdot 6 - 6^2 = 75$. A nonprincipal adjacency eigenvalue θ maps to

$$3 - 2\theta - \theta^2 = 4 - (\theta + 1)^2.$$

Applying this to (10) gives the claimed distance spectrum. Finally, regularity gives $\delta^*(R) = 6$, completing the proof. $\square$

6. Smaller induced counterexamples

The 40-vertex graph contains strict counterexamples of orders 39 and 38. The accompanying programs verify the finite claims in this section using exact arithmetic; no theorem relies on floating-point eigenvalues.

Deleting the endpoints of an edge. Delete the adjacent vertices

$$a = P_{1,0}, \qquad b = P_{1,1}$$

from R, and call the resulting graph H. A graph6 string in this fixed labeling, together with complete adjacency lists and an edge list, is provided in `data/graphs/`.

Theorem 6.1. *The graph H is connected, has 38 vertices, 109 edges, girth five, diameter three, and degree multiset $6^{(28)}, 5^{(10)}$. Moreover,*

$$\delta^*(H) = \frac{17}{3}, \qquad \partial_{38}(H) = -3 - \sqrt{7}.$$

Consequently

$$\delta^*(H) + \partial_{38}(H) = \frac{8}{3} - \sqrt{7} > 0,$$

so H is a strict counterexample to WOW-284.

Proof. Induced deletion preserves simplicity and cannot create a triangle or a 4-cycle. A distinct P-layer remains a 5-cycle. Exact breadth-first search from every vertex gives connectedness and diameter three.

Put

$$X = N_R(a) \setminus \{b\}, \qquad Y = N_R(b) \setminus \{a\}.$$

The girth condition implies that X and Y are disjoint and that no vertex of $X \cup Y$ has a remaining neighbor in $X \cup Y$. These ten vertices therefore have degree five and all their remaining neighbors have degree six; every other vertex has degree six. The handshake lemma now gives

$$|E(H)| = \frac{28 \cdot 6 + 10 \cdot 5}{2} = 109.$$

Each of the ten degree-five vertices has five degree-six neighbors and hence dual degree 6. A degree-six vertex has at most one neighbor in X and at most one in Y, since two in either set would create a 4-cycle. If it has $t \in \{0, 1, 2\}$ neighbors in $X \cup Y$, then

$$d^*(v) = \frac{5t + 6(6 - t)}{6} = 6 - \frac{t}{6} \geq \frac{17}{3}.$$

There are $10 \cdot 5 = 50$ incidences from $X \cup Y$ to the 28 degree-six vertices, so some degree-six vertex has $t = 2$. The lower bound is therefore attained, and $\delta^*(H) = 17/3$.

The exact distance characteristic polynomial is displayed in Appendix A. It contains $(x^2 + 6x + 2)^2$, whose roots are $-3 \pm \sqrt{7}$. For the square-free part of this polynomial, the exact Sturm sequence has variation counts

$$V(-\infty) = 26, \qquad V(-28/5) = 25.$$

Since $P_{38}(-28/5) \neq 0$, Sturm's theorem therefore shows that P_{38} has exactly one distinct root in $(-\infty, -28/5)$. Moreover, $7 > (13/5)^2$, so

$$-3 - \sqrt{7} < -3 - \frac{13}{5} = -\frac{28}{5}.$$

Therefore this is the least distance eigenvalue. Independently, an exact rational LDL^{T} decomposition of $3D(H) + 17I$ has 38 positive pivots. Finally $8/3 > \sqrt{7}$, because $64 > 63$, proving strictness. □

Orders 39 and 42. For $v \in V(R)$, put $H_v = R - v$. Deleting v leaves its six neighbors with degree five and the other 33 vertices with degree six. The six degree-five vertices are pairwise nonadjacent, and all five remaining neighbors of each have degree six; hence their dual degree is 6. A degree-six vertex has at most one degree-five neighbor, since two would form a 4-cycle through v. There are $6 \cdot 5 = 30$ incidences between the degree-five and degree-six vertices. Therefore exactly 30 degree-six vertices have one degree-five neighbor, and

$$\delta^*(H_v) = \frac{5 + 5 \cdot 6}{6} = \frac{35}{6}.$$

For the explicit representative

$$H_{39} := R - P_{1,0},$$

an exact rational LDL^{T} decomposition of $6D(H_{39}) + 35I$ has all 39 pivots strictly positive. Thus $6D(H_{39}) + 35I \succ 0$, so H_{39} is a strict counterexample. Proposition 6.2 below verifies the corresponding assertion for every choice of v.

For another regular example, let X_{42} be the graph induced by the vertices at distance two from $P_{0,0}$ in the Hoffman–Singleton graph. The graph X_{42} is 6-regular and has

$$\mathrm{Spec}(A(X_{42})) = \{6^{(1)}, 2^{(21)}, (-1)^{(6)}, (-3)^{(14)}\},$$
$$\mathrm{Spec}(D(X_{42})) = \{81^{(1)}, 4^{(6)}, 0^{(14)}, (-5)^{(21)}\}.$$

Hence $\delta^*(X_{42}) = 6 > 5 = -\partial_{42}(X_{42})$. Section 8 proves the general construction. The adjacency spectrum is also tabulated by van Dam and Haemers [10, Table 3].

Exact labelled descendant families.

Proposition 6.2. *For every vertex v of R, the graph $R - v$ is connected, has girth at least five, has $\delta^* = 35/6$ and the same exact distance characteristic polynomial P_{39} displayed in Appendix A. Every one of these 40 labelled graphs is a strict counterexample.*

For every edge uv of R, the graph $R - \{u, v\}$ is connected, has girth at least five, has $\delta^* = 17/3$, the same exact distance characteristic polynomial P_{38}, and least distance eigenvalue $-3 - \sqrt{7}$. Every one of these 120 labelled graphs is a strict counterexample.

Proof. The script `scripts/verify_descendant_families.py` enumerates all 40 vertices and all 120 edges. For every induced graph, breadth-first search from every vertex checks connectivity and reconstructs the integer distance matrix. Induced deletion cannot create a cycle of length below five. The script then computes every dual degree as a rational number and the characteristic polynomial $\det(xI - D)$ over $\mathbb{Z}$. Exact equality with P_{39} or P_{38} proves the common-polynomial assertions. The representative H_{39} satisfies $6D(H_{39}) + 35I \succ 0$, so every root of P_{39} is greater than $-35/6$. Since every graph $R - v$ has this same characteristic polynomial, the same strict bound holds for all 40 labelled deletions. An exact Sturm count for P_{38} identifies its least root as $-3 - \sqrt{7}$. No graph-isomorphism or transitivity assumption is used. □

Remark 6.3. A numerical screen of all $\binom{40}{3} = 9880$ three-vertex deletions found no graph with $\Phi > 0$; its best value was approximately -0.306979936667. This is exploratory evidence only, not an exact elimination of order 37 and not a minimality proof.

7. A diameter-three spectral criterion

Theorem 7.1. *Let G be a connected k-regular graph on n vertices, of girth at least five and diameter three. Then*

$$D = 3J + (k - 3)I - 2A - A^2. \tag{12}$$

The principal distance eigenvalue is $3n - k^2 - k - 3$, and every nonprincipal adjacency eigenvalue θ gives the distance eigenvalue

$$\mu(\theta) = k - 2 - (\theta + 1)^2.$$

Consequently G is a strict counterexample to WOW-284 precisely when

$$\max_{\theta \neq k} |\theta + 1| < \sqrt{2k - 2}.$$

Proof. The absence of triangles and 4-cycles means that two vertices at distance two have exactly one common neighbor. Hence the distance-two matrix is $A_2 = A^2 - kI$. Since the diameter is three, $A_3 = J - I - A - A_2$. Substitution into $D = A + 2A_2 + 3A_3$ proves (12). The asserted eigenvalues follow by applying the identity to $\mathbf{1}$ and to $\mathbf{1}^\perp$. The matrix D is nonnegative and irreducible (indeed, every off-diagonal entry is positive), and its row sums are constant by the displayed identity. Perron–Frobenius therefore makes the principal eigenvalue its largest eigenvalue, so the least distance eigenvalue occurs among the nonprincipal images. Finally regularity gives $\delta^* = k$, so strict violation is equivalent to $k > \max_{\theta \neq k}((\theta + 1)^2 - k + 2)$, which is the displayed criterion. □

Thus, in diameter three, WOW-284 is governed by the spread of the nonprincipal adjacency spectrum about -1.

Corollary 7.2 (Bipartite obstruction)**.** *A connected k-regular bipartite graph of diameter three and girth at least five is not a strict counterexample for $k \geq 3$.*

Proof. The nonprincipal adjacency eigenvalue $-k$ maps to

$$\mu(-k) = k - 2 - (k - 1)^2.$$

Its negative is at least k for $k \geq 3$, with equality only at $k = 3$. □

For comparison, negative controls follow directly from Theorem 7.1. The Odd graph $O_4 = KG(7, 3)$ has adjacency spectrum

$$\{4^{(1)}, 2^{(14)}, (-1)^{(14)}, (-3)^{(6)}\},$$

so $\delta^* = 4$ and $\partial_n = -7$. The Heawood graph has adjacency spectrum

$$\{3^{(1)}, (\sqrt{2})^{(6)}, (-\sqrt{2})^{(6)}, (-3)^{(1)}\},$$

so $\delta^* = 3$ and $\partial_n = -2 - 2\sqrt{2}$.

8. Moore second subconstituents

Theorem 8.1. *Let M be a degree-K Moore graph of diameter two, where $K \geq 3$, and fix a vertex v. Let X be the graph induced by the vertices at distance two from v. Then*

$$|V(X)| = K(K-1), \qquad X \text{ is } (K-1)\text{-regular},$$

X has girth at least five and diameter three, and

$$\partial_{K(K-1)}(X) = -\frac{5 + \sqrt{4K-3}}{2}.$$

Consequently

$$\delta^*(X) + \partial_{K(K-1)}(X) = K - 1 - \frac{5 + \sqrt{4K-3}}{2},$$

which is positive exactly for integers $K \geq 6$.

Proof. Since X is an induced subgraph of the girth-five graph M, it contains no triangle or 4-cycle. Hence $g(X) \geq 5$.

Relative to $\{v\} \sqcup N(v) \sqcup V(X)$, write

$$A(M) = \begin{pmatrix} 0 & \mathbf{1}^\top & 0 \\ \mathbf{1} & 0 & C \\ 0 & C^\top & B \end{pmatrix}.$$

All zero and all-one blocks in this proof have the dimensions forced by the displayed partition. Each vertex of X has one neighbor in $N(v)$ and $K-1$ in X, so B is $(K-1)$-regular. Distinct rows of C have disjoint support and row sum $K-1$, hence $CC^\top = (K-1)I_K$. Therefore $C^\top$ is injective and

$$\mathbb{R}^{V(X)} = \operatorname{im} C^\top \oplus \ker C.$$

Moreover,

$$C^\top \mathbf{1}_K = \mathbf{1}_{V(X)}.$$

The middle-right block of the Moore identity gives $CB = J - C$. Transposing this identity shows that

$$BC^\top z = -C^\top z \qquad (z \perp \mathbf{1}_K).$$

Thus B has eigenvalue -1 on the $(K-1)$-dimensional space $C^\top(\mathbf{1}_K^\perp)$, while $\mathbf{1}_{V(X)}$ has eigenvalue $K-1$.

Now let $x \in \ker C$. Since $C^\top \mathbf{1}_K = \mathbf{1}_{V(X)}$,

$$\mathbf{1}_{V(X)}^\top x = \mathbf{1}_K^\top C x = 0.$$

Furthermore,

$$CBx = (J - C)x = 0,$$

so $\ker C$ is B-invariant. The bottom-right block of the Moore identity therefore restricts on $\ker C$ to

$$B^2 + B - (K-1)I = 0,$$

whose roots are

$$r = \frac{-1 + \sqrt{4K-3}}{2}, \qquad s = \frac{-1 - \sqrt{4K-3}}{2}.$$

The kernel has dimension $K(K-2) > 0$, so the principal line, $C^\top(\mathbf{1}_K^\perp)$, and $\ker C$ account for all

$$1 + (K-1) + K(K-2) = K(K-1)$$

dimensions. If the multiplicities of r, s there are m_r, m_s, then the trace of B gives $m_r r + m_s s = 0$, because the principal eigenvalue $K-1$ cancels the $(K-1)$ copies of -1. Since $r > 0 > s$, both multiplicities are positive.

It remains to prove that X has diameter three. Around any $x \in X$, girth at least five gives 1, $K-1$, and $(K-1)(K-2)$ distinct vertices in the first three distance layers of X. The remaining $K-2$ vertices are exactly the other vertices of X sharing the unique neighbor of x in $N(v)$. They have no path of length at most two to x, since such a path would create a triangle or 4-cycle in M. Every neighbor of any such remaining vertex z is nonadjacent to x, since an edge to x would complete the 4-cycle through their shared neighbor in $N(v)$. If $w \in N_X(z)$, the unique ambient common neighbor of x and w cannot lie in $N(v)$: it would have to be the unique such neighbor of x, and would then form a triangle with w, z. It therefore lies in X, giving a path of length three from x to z. Hence all remaining vertices are at distance three. This proves connectivity and diameter three.

Apply Theorem 7.1 with $k = K - 1$. The eigenvalue -1 maps to $K - 3$, while r and s map respectively to

$$-\frac{5 + \sqrt{4K - 3}}{2} \quad \text{and} \quad -\frac{5 - \sqrt{4K - 3}}{2}.$$

Thus the first of these is the least distance eigenvalue. Finally,

$$2\big(\delta^*(X) + \partial_{K(K-1)}(X)\big) = 2K - 7 - \sqrt{4K - 3}.$$

This is nonpositive for $K = 3, 4, 5$. For $K \geq 6$, both sides of $2K - 7 > \sqrt{4K - 3}$ are positive, and

$$(2K - 7)^2 - (4K - 3) = 4(K^2 - 8K + 13) > 0.$$

Indeed, $K^2 - 8K + 13 = 1$ at $K = 6$ and is increasing for $K \geq 6$. Hence the gap is positive exactly for integers $K \geq 6$. □

For $K = 7$, the two roots on $\ker C$ are 2 and -3. Their multiplicities m_2, m_{-3} satisfy

$$m_2 + m_{-3} = 35, \qquad 2m_2 - 3m_{-3} = 0,$$

the second equation following from the trace after the principal eigenvalue 6 cancels the six copies of -1. Hence

$$m_2 = 21, \qquad m_{-3} = 14.$$

This gives the explicit 42-vertex graph above. A degree-57 Moore graph, if it exists, would give a 56-regular example on 3192 vertices. Its existence remains open [5]; the theorem is therefore a parameterized construction, not an unconditional infinite family.

9. An equitable-deletion framework

Proposition 9.1. *Let M be a degree-K Moore graph of diameter two. Partition its vertices as $V(M) = U \sqcup S$, write*

$$A(M) = \begin{pmatrix} B & C \\ C^{\mathsf{T}} & P \end{pmatrix},$$

and suppose $M[S]$ is r-regular while every vertex of U has exactly $t > 0$ neighbors in S. Then $M[U]$ is $(K - t)$-regular. If $Py = \eta y$ and $y \perp \mathbf{1}$, then

$$BCy = -(1 + \eta)Cy, \qquad \|Cy\|^2 = (K - 1 - \eta - \eta^2)\|y\|^2.$$

Moreover, on $\ker C^{\mathsf{T}}$,

$$B^2 + B - (K - 1)I = 0.$$

Whenever $M[U]$ has diameter three, these adjacency data feed directly into Theorem 7.1.

Proof. Each vertex of U has degree K in M and exactly t neighbors in S. It therefore has $K - t$ neighbors in U, so $M[U]$ is $(K - t)$-regular. All zero and all-one blocks in this proof have the dimensions forced by the displayed partition.

The upper-right and lower-right blocks of the Moore identity are

$$BC + CP = -C + J, \qquad C^{\mathsf{T}}C + P^2 = (K - 1)I - P + J.$$

If $Py = \eta y$ and $y \perp \mathbf{1}$, then $Jy = 0$, so the upper-right block gives

$$BCy = -(1 + \eta)Cy.$$

Taking the inner product of the lower-right block with y gives

$$\|Cy\|^2 = (K - 1 - \eta - \eta^2)\|y\|^2.$$

Finally, if $x \in \ker C^{\mathsf{T}}$, then $C\mathbf{1} = t\mathbf{1}$ and $t > 0$ imply $x \perp \mathbf{1}$. The upper-left block consequently gives

$$(B^2 + B - (K-1)I)x = 0.$$

□

For the 40-vertex graph, $U = V(R)$, $K = 7$, S is the induced Petersen graph, $r = 3$, and $t = 1$. If, more generally, $S \subsetneq V(M)$ is itself a degree-r Moore graph and every outside vertex has exactly one neighbor in S, cross-edge counting gives

$$(r^2 + 1)(K - r) = K^2 - r^2.$$

Because the outside set is nonempty, $K > r$, so cancellation yields $K = r^2 - r + 1$. The coordinate deletions displayed above realize the steps $C_5 \subset$ Petersen $\subset$ Hoffman–Singleton, corresponding to $(r, K) = (2, 3)$ and $(3, 7)$, but do not yield an infinite chain.

10. Further exact and exploratory checks

For $i \in \mathbb{F}_5$, put $\mathcal{L}_i = \{P_{i,j}, Q_{i,j} : j \in \mathbb{F}_5\}$, and let R_m be the graph obtained from G by deleting $\mathcal{L}_0, \ldots, \mathcal{L}_{m-1}$ (with $R_0 = G$). These balanced coordinate-layer deletions give the following exact representatives:

m	$\lvert V\rvert$	degree	∂_n	$\delta^* + \partial_n$
0	50	7	−4	3
1	40	6	−5	1
2	30	5	−6	−1
3	20	4	−7	−3
4	10	3	−3	0

Thus this natural chain stops producing strict counterexamples after the 40-vertex member. The preceding analytic theorems cover the Moore and regular diameter-three mechanisms, Moore second subconstituents, and the equitable deletion framework. The exact computations check the listed finite examples, all singleton and edge-endpoint deletions of R, the balanced layer chain, O_4, and the Heawood graph.

No unconditional infinite family is established. Fixed-degree, diameter-three graphs have bounded order by the Moore bound, so an infinite family within this framework would require unbounded degree. Covers, subdivisions, coronas, and general vertex replacements are not asserted to preserve the strict inequality; the present arguments establish no such preservation theorem.

11. Computational certificate

The repository contains independent exact checks. For the 50-vertex graph, the original certificate does the following:

(1) constructs the 175 unordered edges and checks all 50 degrees;
(2) checks the common-neighbor count for every unordered vertex pair;
(3) constructs all 2,500 distance entries by integer breadth-first search;
(4) computes the distance characteristic polynomial symbolically; and
(5) computes an exact rational $L\Delta L^{\mathsf{T}}$ decomposition of $D + 7I$ and checks that all 50 pivots are positive.

The resulting characteristic polynomial is

$$(t - 91)(t - 1)^{21}(t + 4)^{28},$$

and all 50 rational pivots are positive. Set

$$M := D + 7I.$$

The exact decomposition is

$$M = L\Delta L^{\mathsf{T}},$$

where L is unit lower triangular and Δ has strictly positive diagonal. Hence L is invertible, and for every nonzero y,

$$y^{\mathsf{T}} M y = (L^{\mathsf{T}} y)^{\mathsf{T}} \Delta (L^{\mathsf{T}} y) > 0.$$

Thus $M = D + 7I$ is positive definite.

A second script constructs R by the stated deletion and independently checks its 40 degrees, all BFS distances, girth, the block identities (8)–(9), and the characteristic polynomials

$$\det(tI - B) = (t-6)(t-2)^{18}(t-1)^4(t+2)^5(t+3)^{12},$$
$$\det(tI - D(R)) = (t-75)(t-3)^5 t^{16}(t+5)^{18}.$$

The extended verifier reconstructs the 42-, 40-, 39-, and 38-vertex graphs from the field-coordinate rules and checks all structural, dual-degree, characteristic-polynomial, Sturm, and rational $LDL^{\top}$ assertions used above. A separate 38-vertex audit starts from the stored graph6 representation, reconstructs graph distances by integer breadth-first search through NetworkX [11], and uses a handwritten `Fraction`-based decomposition. This audit uses a different graph representation and an independent $LDL^{\top}$ implementation; the underlying graph is the same.

Finally, `scripts/verify_descendant_families.py` exhausts the 40 labelled single-vertex and 120 labelled edge-endpoint deletions. SymPy supplies exact integer characteristic polynomials $\det(tI - D)$ and Sturm counts of their square-free parts [12]. The recorded output is `supplement/extended_2026-07-23/logs/descendant_family_output.txt`; the enclosing `SHA256SUMS` records the expected SHA-256 digest of that output and every other archived certificate, permitting byte-for-byte integrity checks. The cross-platform command

```
python scripts/verify_extended.py
```

runs the exact certificates for the explicit non-50 constructions, the independent graph6 audit, the descendant-family verification, and the byte-for-byte graph-export checks. The balanced layer-deletion chain, the Odd graph, and the Heawood graph are checked by

```
python scripts/explore_generalizations.py
```

which also performs the separately labelled floating-point three-vertex screen. No floating-point eigenvalue evidence is used for any theorem.

12. Formal verification

The explicit 50-vertex counterexample is fully formalized and verified in Lean 4.31 with Mathlib 4.31 [13, 14]. The development verifies the graph, 7-regularity, the exhaustive common-neighbor certificate, girth five, and the adjacency-square and distance-matrix identities. It supplies exact rational matrices S, S^{-1}, proves their two-sided inverse identities by denominator-cleared integer certificates, proves $DS = S\,\mathrm{diag}(91, (-4)^{(28)}, 1^{(21)})$, and checks the multiplicities. Thus the complete explicit 50-vertex proof, including its least distance eigenvalue, is kernel-checked. The scalar $k \le 3$ threshold is formalized separately.

Lean 4.31 also kernel-checks finite spectral certificates for the explicit 38-, 39-, 40-, and 42-vertex constructions. For orders 38, 39, and 42, the public endpoints certify the exact minimum dual-degree data, positive definiteness of a shifted finite matrix, and the resulting strict inequality for every nonzero real eigenpair. For order 40, the endpoint certifies a two-sided invertible exact diagonalization, its diagonal-entry multiplicities, and that -5 is an attained minimum diagonal entry, together with dual degree six and gap one.

The scope of these non-50 results is deliberately finite and spectral: their public endpoints do not bundle every structural hypothesis, and the semantic finite matrices are not identified with Mathlib's `SimpleGraph.dist` in a single theorem. They are therefore described here as kernel-checked finite spectral certificates, rather than as a full standard-library formalization of every graph-theoretic phrase in the manuscript. Section 1, the diameter-three criterion, and the Moore-subconstituent theorem remain conventional analytic proofs. The descendant-family assertions are conventional finite arguments supported by exact computation.

Scope, provenance, and disclosures

The note gives counterexamples of orders 38, 39, 40, 42, and 50. No claim is made that 38 is the minimum possible order among WOW-284 counterexamples, and no exhaustive search over all smaller graphs is part of this note. Calling the 40-vertex graph the $(6,5)$-cage refers to its classical minimum-order property among 6-regular girth-five graphs, not to minimum order for the WOW problem. The distance-spectral input for Moore graphs is credited to Howlader and Panigrahi; the classical 40-vertex cage and 42-vertex second subconstituent are separately credited above. The repository source ledger records the explicit WOW-284 connection and the limits of the targeted priority search.

OpenAI ChatGPT-5.6 Sol Pro assisted with adversarial proof checking, proof exploration, and Lean formalization. No AI system is an author. Samuil Petkov is the sole named author and assumes full responsibility for the mathematics, citations, attribution, and conclusions.

The author received no funding for this work and declares no competing interests. The manuscript source, exact verification code, machine-readable certificates, and build instructions are available in the public repository `github.com/SamPetkov/wow284`. This manuscript corresponds to GitHub release `v2.1.0`. Its contents are identified by the repository manifest and `SHA256SUMS`.

Appendix A. Exact distance characteristic polynomials

For every labelled single-vertex deletion considered in Proposition 6.2,

$$\begin{aligned} P_{39}(x) = {} & x^9(x+5)^{12}(x^2+6x+3) \\ & \cdot (x^3+3x^2-15x-7)^2(x^3+3x^2-15x-3)^2 \\ & \cdot (x^4-78x^3+303x^2-70x-450). \end{aligned}$$

For every labelled edge-endpoint deletion considered there,

$$\begin{aligned} P_{38}(x) = {} & x^4(x-2)(x+5)^8(x^2+6x+2)^2 \\ & \cdot (x^3+3x^2-15x-3) \\ & \cdot (x^4+5x^3-7x^2-23x-6) \\ & \cdot (x^5+9x^4+7x^3-77x^2-54x-4) \\ & \cdot (x^9-67x^8-404x^7+1772x^6+7205x^5 \\ & \qquad -18489x^4-17018x^3+20288x^2+16824x+1680). \end{aligned}$$

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